# Where do libraries come from, and where do they stop helping?

## Bootstrapped libraries, documentation, evidence selection, and boundaries

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### In one sentence

A 4-bit 8B model wrote a verified library for a new benchmark from three examples, and a 0.6B caller that never saw the benchmark uses it correctly to 1M tokens; the useful capacity is in the proposer and in choosing evidence, not in readers, recursion or larger controllers; a retrieval library also beats direct reading on LongBench-v2.

# Libraries can be bootstrapped: an 8B wrote MRCR's library.

| MRCR held out (8k to 1M tokens) | 2 needles | 4 needles |
|---|---|---|
| Direct answer, 4B | 0.23 | 0.15 |
| RLM, untrained 4B, no library | 0.18 | 0.11 |
| 8B-written library, run alone | 1.000 | 1.000 |
| 0.6B caller + library | **1.000** | **1.000** |
| Untrained 4B + library | 0.95 | 0.88 |

64 rows per column. The 0.6B caller never saw MRCR. The proposer saw the contract and three solved examples; candidates run on 30 teaching tasks; the best scored 30/30 and was reviewed before use (40 lines of parsing, nothing copied from the examples).

## Who can write it (seeds accepted)

4B Instruct: 0 of 1
4B, thinking off / on: 0 of 1 / 0 of 2
8B, thinking off: 0 of 2
8B, thinking on: **2 of 5**
14B (4-bit), thinking on: **2 of 5**
4B after SFT + RL: 0 of 4; test-time RL on MRCR: 0 in 1,920 samples

Caveat: MRCR is solvable by a short parser, so this measures program discovery, not reading. We found no published note saying so.

Ledger E21, E21b; runs e21-*. MRCR: openai/mrcr (MIT), official SequenceMatcher grader.

# Discovery on 37 generated families: thinking, scale, and training.

| Proposer | Solved / 37 | Tokens |
|---|---|---|
| 4B Instruct (2 seeds) | 5, 4 | 4k |
| 8B, no thinking | 4 | 4k |
| 4B, thinking | 14 | 55k |
| 8B, thinking (2 seeds) | 20, 25 | 48k |
| 14B, thinking (2 seeds) | 21, 26 | 30k |
| 4B + SFT (2 seeds) | 9, 7 | 3k |
| 4B + SFT + RL (2 seeds) | 16, 14 | 2k |
| 4B, round four (2 seeds) | **22, 20** | 2k |
| avg. of 4 RL seeds | 15.8 | |
| 4B, 3x helper data, avg. | 16.8 | 2k |

Tokens: proposer output per family. Families vary record format, question and answer format; held-out families use a new format or question.

**Training the proposer:** SFT on the 8B's verified code, then 40 steps of RL with the verified teaching score as reward: 9 to 30 of 74 attempts ($p < 0.001$). Seed-sensitive: round four's 42 (8B 45) is the best of four RL seeds (22 to 42). MRCR still fails.

**Helper gap:** 3x the helper teacher data matches round four on solves, reaches the 8B on helper families (mean 0.46), and varies far less by seed.

Ledger E26 to E26g, E36 to E46; runs e26-disc-*, e4*. Generator: family_gen.py (answers exact).

# Bootstrapping is cheap where answers are exact (pre-registered).

| 30 unseen families, 2 seeds | Correct | False acc. | Tokens |
|---|---|---|---|
| 4B, best of 8, 3 labels | **12, 14** | 0, 1 | 3.7k |
| 8B thinking, 30 labels | 13, 16 | 0, 0 | 52.4k |

Correct = accepted library scores 0.9+ on 30 fresh tasks with twice the records. Pooled 26 vs 29 of 60, p 0.61. Protocol, predictions and analysis committed before the runs; met in both seeds and for all five trained 4B checkpoints (12 to 16 vs 13).

## Why three labels suffice

Candidates that run cleanly and reproduce the prompt's examples are correct: precision 0.81, 0.88, 0.98 with 1, 2, 3 labels (127 of 129 at three, all generalizing). Agreement among candidates is no substitute: an RL-trained proposer repeats its wrong programs.

Limit: on noisy families (MuSiQue F1) ranking close routines takes tens of labels.

Ledger E37, E38, E39; runs e37*, e39*; docs/notebook/assets-e39-preregistration.md.

# One line of documentation decides whether an untrained caller uses a library.

| MRCR, untrained 4B caller, docstring | Score |
|---|---|
| Signature only | 0.18 |
| Description + signature (as the 8B wrote it) | 0.59 |
| Signature + usage line | 0.95 |
| Description + signature + usage line | 0.95 |
| Hand-written library, usage line removed | 0.22 |
| Hand-written library, as written | 0.95 |

Usage line: `Typical use: FINAL_TEXT(answer_mrcr(request)).`

Without it the untrained 4B hands the whole input to a sub-model (overflowing at 128k and above) or retypes the routine's output by hand.

**The distilled 0.6B scores 1.000 without the line; the untrained 0.6B fails either way (0.21, 0.13).** Robustness comes from distillation, not size. The bootstrapper now inserts the line.

Ledger E25c, E25d; runs e25c-*, e25d-*. One family, one library, one seed per arm.

# MuSiQue in long haystacks: capacity belongs in choosing evidence.

| Answer F1 (90 to 180 held-out questions per row) | 8k | 32k | 128k | tokens |
|---|---|---|---|---|
| Direct answer, 4B | 0.52 | 0.25 | overflow | |
| RLM without a library, depth 0 / depth 1 | 0.49 / 0.51 | 0.25 / 0.25 | overflow | |
| Library that hands each hop to a child RLM | 0.29 | 0.29 | 0.34 | 3 to 4 times |
| 0.6B caller + filter, BM25, read | 0.52 | 0.31 | 0.46 | 57k |
| 0.6B caller + two-round retrieval | **0.53** | **0.39** | **0.50** | 10k |

**What helps:** a second, targeted retrieval (BM25 for the question, the helper names the missing fact, BM25 again, one read): +0.058 over BM25 alone, +0.045 over filter+BM25 at a sixth of the tokens.

**What does not:** recursion; an 8B reader (+0.01; gold paragraphs reach 0.63 to 0.67 with either reader); more sampling (majority of 3 at 88k tokens: 0.22 vs 0.35 for the library at 32k).

Ledger E22c to E22f, E23, E25e, E28. MuSiQue (CC BY 4.0) dev questions in haystacks of other questions' paragraphs.

# External validity, and what we withdrew.

## LongBench-v2 (all 503 questions, chance 0.25)

Direct (truncated): 0.356, 42k tokens
**Retrieve-then-read: 0.427, 10k tokens**
via the 0.6B caller: 0.423 (+0.07, se 0.02)

Selecting evidence beats summarizing (map-reduce: no gain).

**Boundary:** no proposer bootstraps OOLONG, our own family (8B 0.20, 14B 0.00; shipped library 0.60): 30 examples span 11 question templates.

**Withdrawn:** distractor-limited reader; length routing (pre-registered); "no help on LongBench-v2" (it was map-reduce).

**Method lessons**

- Verify under the semantics you run under: the kernel's `ask` raises on a cut-off reply, scripted harnesses returned partial text (one routine crashed in 46 of 90 caller runs).
- Verification must span deployment lengths: 8k-only teaching selected a pass-through routine that scores 0 at 128k.
- Routine evolution (GEPA-style): prompt rewrites by the 8B hurt; numbers helped (+0.048, ns); random search found nothing.

Ledger E32 to E32e, E31, E22b, E24, E24c, E35. LongBench-v2: zai-org (Apache-2.0).

## What this changes, and what to test next.

**The thesis, sharpened.** For small recursive models, capability lives outside the weights: verified routines (bootstrapped by a capable proposer, or by a small one trained on verified code and rewarded by verification; on exact families three labels suffice), documentation a thin caller can act on, and evidence selection inside the routine. A 0.6B caller with a retrieval library beats a 4B reading the document on MuSiQue and on LongBench-v2.

**Next**

1. Proposer RL is seed-sensitive (22 to 42 of 74 for one recipe): compare recipes over 3+ RL seeds, and select checkpoints by held-out discovery, not training reward.
2. Labels on noisy families: graded verification for MuSiQue- and OOLONG-like families.
3. Breadth: split a broad family into question templates automatically (OOLONG: counting yes, timeline and user no).
4. A short note on program-solvable long-context benchmarks (MRCR; candidates: Michelangelo, RULER).

# Appendix: protocol and provenance.

Models: Qwen3-4B-Instruct-2507 (helper, caller), Qwen3-0.6B + LoRA (distilled caller), Qwen3-4B hybrid and Qwen3-8B-AWQ (proposers), all served with vLLM 0.27.1 on two A100 MIG 7g.40gb slices.

Data: OpenAI MRCR (MIT; within-bin split, 30 teaching rows, 64 held out per needle count); MuSiQue v1.0 (CC BY 4.0; teaching from train, held out from dev; two samples of 90); LongBench-v2 (Apache-2.0, revision 2b48e494); generated families (family_gen.py, exact answers).

Statistics: paired differences with standard errors over all questions; exact McNemar for family counts; the routing test was pre-registered.

Every number above has a ledger row in docs/RESULTS.md and a run directory in the run store; the lab notebook (docs/notebook/2026-10-19-...) records failures and corrections in time order.